Multivariate Regression Model

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The variance of the OLS Estimators

Gauss Markov assumptions in MLR:

- MLR.1: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_k x_k + u$
- MLR.2: random sampling from the population
- MLR.3: no perfect collinearity in the sample
- MLR.4: $E(u|x_1, ..., x_k) = E(u) = 0$ (exogenous explanatory variables)
- MLR.5: $Var(u|x_1, ..., x_k) = Var(u) = \sigma^2$ (homoskedasticity)

The expected value of the OLS Estimators

- Therefore, the bias in $\tilde{\beta}_1$ is:



$$Bias(\tilde{\beta}_1) = \beta_2 \tilde{\delta}_1$$

Called the **omitted variable bias**

- The simple regression estimator is unbiased in two cases:
  - $\beta_2 = 0$. But this means that x_2 does not appear in the population model.
  - $Corr(x_{i1}, x_{i2}) = 0$. Then the simple and multiple regression estimators are identical because $\tilde{\delta}_1 = 0$.
- Summary of bias in the simple regression estimator of $\tilde{\beta}_1$.

	$Corr(x_1, x_2) > 0$	$Corr(x_1, x_2) < 0$
$\beta_2 > 0$	Positive Bias	Negative Bias
$\beta_2 < 0$	Negative Bias	Positive Bias

The variance of the OLS Estimators

- Assumptions MLR.1 through MLR.5 are called the **Gauss Markov assumptions.**

- Assumptions MLR.1 and MLR.4 imply

$$E(y|x_1, x_2, ..., x_k) = \beta_0 + \beta_1 x_1 + ... + \beta_k x_k$$

- and when we add MLR.5,

$$Var(y|x_1, x_2, ..., x_k) = Var(u|x_1, x_2, ..., x_k) = \sigma^2$$

The variance of the OLS Estimators

- Theorem 3.2 (Sampling variances of the OLS slope estimators)

  Under assumptions MLR.1 – MLR.5:

$$Var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1 - R_j^2)}, \quad j = 1, \ldots, k$$

Variance of the error term

Total sample variation in explanatory variable x_j:

$$\sum_{i=1}^{n} (x_{ij} - \bar{x}_j)^2$$

R-squared from a regression of explanatory variable x_j on all other independent variables (including a constant)

The variance of the OLS Estimators

- **Variances of the OLS slope esimators (Chapter 3 Appendix)**

  In the model with *k* independent variables, we can write $\hat{\beta}_j$ as:

$$\hat{\beta}_j = \beta_j + \frac{\sum_{i=1}^{n} \hat{r}_{ij} u_i}{\sum_{i=1}^{n} \hat{r}_{ij}^2}$$

  where $\hat{r}_{ij}$ is the residual from a regression of x_j on $x_1, \ldots, x_{j-1}, x_{j+1}, \ldots, x_k$

$$Var(\hat{\beta}_j) = Var\left(\frac{\sum_{i=1}^{n} \hat{r}_{ij} u_i}{\sum_{i=1}^{n} \hat{r}_{ij}^2}\right) = \frac{\sum_{i=1}^{n} \hat{r}_{ij}^2 Var(u_i)}{(\sum_{i=1}^{n} \hat{r}_{ij}^2)^2}$$

$$= \frac{\sigma^2}{\sum_{i=1}^{n} \hat{r}_{ij}^2} = \frac{\sigma^2}{SSR_j} = \frac{\sigma^2}{SST_j(1 - R_j^2)}$$

The variance of the OLS Estimators

- **Components of OLS Variances:**
- **1) The error variance**
  - A high error variance increases the sampling variance because there is more “noise” in the equation
  - A large error variance necessarily makes estimates imprecise
  - The error variance does not decrease with sample size
- **2) The total sample variation in the explanatory variable**
  - More sample variation leads to more precise estimates
  - Total sample variation automatically increases with the sample size
  - Increasing the sample size is thus a way to get more precise estimates

The variance of the OLS Estimators

- **3) Linear relationships among the independent variables**

R_j^2 : Regress x_j on all other independent variables (including a constant)

The R-squared of this regression measures how linearly related x_j is to the other explanatory variables.

- Sampling variance of $\hat{\beta}_j$ will be the higher the better explanatory variable x_j can be linearly explained by other independent variables
- The problem of almost linearly dependent explanatory variables is called **multicollinearity** (i.e. $R_j \to 1$ for some j)
- Multicollinearity can cause problems for getting a imprecise estimate

$$R_j^2 \to 1,\ Var(\hat{\beta}_j) \to \infty$$

The variance of the OLS Estimators

- **Theorem 3.1 (Unbiasedness of OLS)**

$$MLR.1-MLR.4 \Rightarrow E(\hat{\beta}_j) = \beta_j, \quad j = 0, 1, \ldots, k$$

Leaving a variable out when it should be included in multiple regression is called **underspecifying the model**, which leads to **omitted variable bias.**

- **Theorem 3.2 (Sampling variances of OLS slope estimators)**

$$MLR.1-MLR.5 \Rightarrow Var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1-R_j^2)}, \quad j = 1, \ldots, k$$

The problem of almost linearly dependent explanatory variables is called **multicollinearity** (i.e. $R_j \to 1$ for some j)

The variance of the OLS Estimators

- Estimating the error variance

$$\hat{\sigma}^2 = \left(\sum_{i=1}^{n} \hat{u}_i^2 \right) / [n - k - 1]$$

An unbiased estimate of the error variance can be obtained by substracting the number of estimated regression coefficients from the number of observations. The number of observations minus the number of estimated parameters is also called the degrees of freedom. The n estimated squared residuals in the sum are not completely independent but related through the k+1 equations that define the first order conditions of the minimization problem.

$$MLR.1 - MLR.5 \quad \Rightarrow \quad E(\hat{\sigma}^2) = \sigma^2$$

- Theorem 3.3 (Unbiased estimator of the error variance)

The variance of the OLS Estimators

- Estimation of the sampling variances of the OLS estimators

The true sampling variation of the estimated β_j

$$sd(\hat{\beta}_j) = \sqrt{Var(\hat{\beta}_j)} = \sqrt{\sigma^2 / \left[SST_j(1 - R_j^2)\right]}$$

Plug in $\hat{\sigma}^2$ for the unknown σ^2

The estimated sampling variation of the estimated β_j

$$se(\hat{\beta}_j) = \sqrt{\widehat{Var}(\hat{\beta}_j)} = \sqrt{\hat{\sigma}^2 / \left[SST_j(1 - R_j^2)\right]}$$

- Note that these formulas are only valid under assumptions MLR.1-MLR.5 (in particular, there has to be homoskedasticity)

The variance of the OLS Estimators

- **How to find** $SST, SSE, SSR,$ and $\hat{\sigma}$ **in Stata?**

  Example: wage equation

  Data: WAGE1.dta

$$lwage = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 tenure + u$$

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. reg lwage educ exper tenure
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Source	SS	df	MS
Model	46.8741776	3	15.6247259
Residual	101.455574	522	.194359337
Total	148.329751	525	.28253286

Number of obs	= 526
F(3, 522)	= 80.39
Prob > F	= 0.0000
R-squared	= 0.3160
Adj R-squared	= 0.3121
Root MSE	= .44086

lwage	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
educ	.092029	.0073299	12.56	0.000	.0776292	.1064288
exper	.0041211	.0017233	2.39	0.017	.0007357	.0075065
tenure	.0220672	.0030936	7.13	0.000	.0159897	.0281448
_cons	.2843595	.1041904	2.73	0.007	.0796756	.4890435

Efficiency of OLS: The Gauss-Markov Theorem

- **Efficiency of OLS: The Gauss-Markov Theorem**
  - Under assumptions MLR.1 - MLR.5, OLS is unbiased
  - However, under these assumptions there may be many other estimators that are unbiased
  - Which one is the unbiased estimator with the smallest variance?
  - In order to answer this question one usually limits oneself to linear estimators, i.e. estimators linear in the dependent variable

$$\tilde{\beta}_j = \sum_{i=1}^{n} w_{ij} y_i$$

May be an arbitrary function of the sample values of all the explanatory variables; the OLS estimator can be shown to be of this form

Efficiency of OLS: The Gauss-Markov Theorem

- **Theorem 3.4 (Gauss-Markov Theorem)**

  Under assumptions MLR.1 - MLR.5, the OLS estimators are the **best linear unbiased estimators (BLUEs)** of the regression coefficients, i.e.

- **E (estimator):** We must be able to compute an estimate from the observable data, using a fixed rule.

- **L (linear):** The estimator is a linear function of $\{y_i : i = 1, 2, ..., n\}$. The OLS estimators can be written in this form:

$$\tilde{\beta}_j = \sum_{i=1}^{n} w_{ij} y_i$$

  where the w_{ij} are any functions of $\{(x_{i1}, ..., x_{ik}) : i = 1, ..., n\}$.

Efficiency of OLS: The Gauss-Markov Theorem

- **U (unbiased):** We must impose enough restrictions on the w_{ij}, so that

$$E(\tilde{\beta}_j) = \beta_j, \; j = 0, 1..., k$$

the OLS estimators are unbiased under MLR.1 through MLR.4

- **B (best):** Are the OLS estimators the **unbiased estimator** with the smallest variance? In other words, under MLR.1 through MLR.5, and conditional on the explanatory variables in the sample:

$$Var(\hat{\beta}_j) \leq Var(\tilde{\beta}_j) \text{ all } j$$

OLS is only the best estimator if MLR.1 – MLR.5 hold; if there is heteroscedasticity for example, there are better estimators.

Efficiency of OLS: The Gauss-Markov Theorem

- **Discussion of the Gauss-Markov Theorem**
  - If the Gauss-Markov assumptions hold, and we insist on **unbiased** estimators that are also **linear** functions of dependent variable, then we need look no further than OLS: it delivers the **smallest** possible variances.
  - It might be possible (but not practical) to find unbiased estimators that are nonlinear functions of dependent variable that have smaller variances than OLS.
  - Failure of MLR.5 does not cause bias in the $\hat{\beta}_j$, but it has two consequences:
    1) The formulas for $Var(\hat{\beta}_1)$ and $se(\hat{\beta}_1)$ are wrong
    2) The $\hat{\beta}_j$ are no longer BLUE

Conclusion

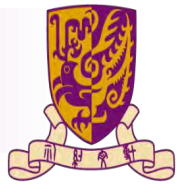
- **Conclusion:**
  - Motivation for multiple Regression
  - Mechanics and interpretation of OLS
  - The expected value of the OLS Estimators
  - The variance of the OLS Estimators
  - Efficiency of OLS: The Gauss-Markov Theorem

Review of Statistics

The framework allows rigorous statistical inference about unknown parameters (unknown numbers) of population distribution using a sample of data from that population, which is the main goal of statistics

- Estimation
- Hypothesis testing
- Confidence interval

Review of Statistics



Estimation

- Calculate a single value that serves as a "guess" for some unknown parameter of the population distribution. It is a function of $Y_1, \ldots, Y_n$, called estimator.
  - For example, $\bar{Y}$ can serve as an estimator for μ_Y. What are properties of $\bar{Y}$? Why not other estimators?
  - More generally, how to get a "best guess" or best estimator?

Sampling distribution

- The distribution of an estimator. Note that estimator is a function of $Y_1, \ldots, Y_n$, so it is a random variable
  - In general, it depends on n and population distribution
  - What is the sampling distribution of $\bar{Y}$?

Review of Statistics

Example: Suppose Y takes on 0 or 1 (a Bernoulli random variable) with the distribution,

$$P(Y = 0) = 0.22 = 1 - p, P(Y = 1) = 0.78 = p\ .$$

Then $E[Y] = p = 0.78$ and

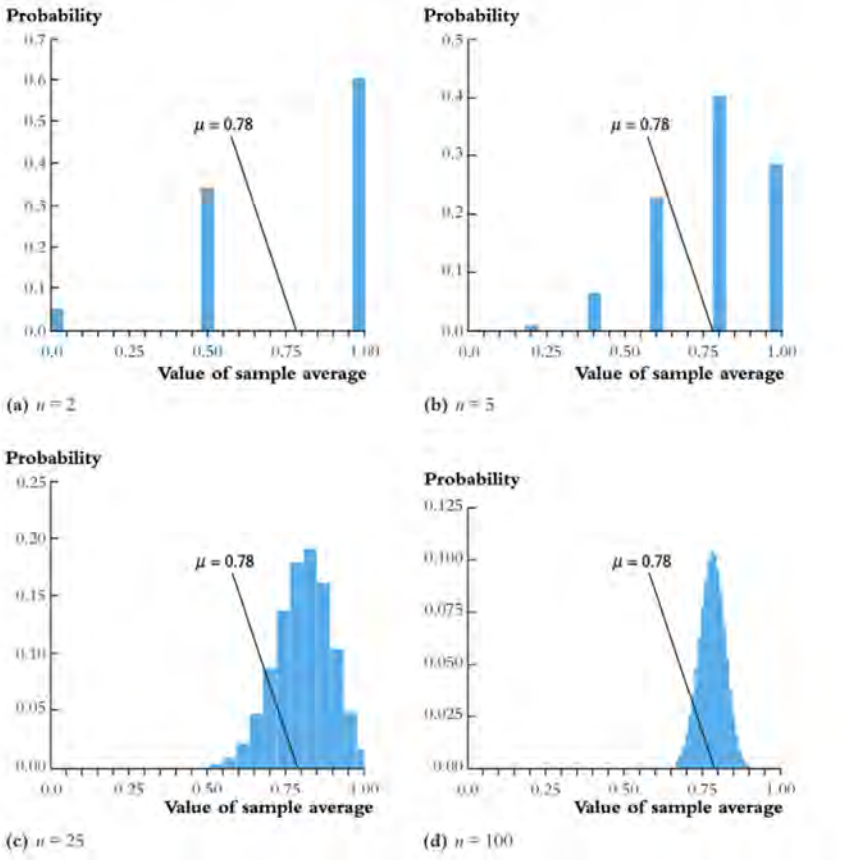
$$var(Y) = p(1 - p) = 0.1716 \text{ (remember this?)}$$

Consider $n = 2$. $\bar{Y}$ can take $0, 1/2$ and 1. The sampling distribution of $\bar{Y}$ is

$$P(\bar{Y} = 0) = 0.22^2 = 0.0484$$
$$P(\bar{Y} = 1/2) = 2 \times 0.22 \times 0.78 = 0.3432$$
$$P(\bar{Y} = 1) = 0.78^2 = 0.6084$$

Review of Statistics

The sampling distribution of $\bar{Y}$ when Y is Bernoulli ($p = 0.78$)



Review of Statistics

- $\bar{Y}$ is an unbiased estimator for μ_Y, since

$$E[\bar{Y}] = E[\frac{1}{n}\sum_{i=1}^{n} Y_i] = \frac{1}{n}\sum_{i=1}^{n} E[Y_i] = \frac{1}{n}\sum_{i=1}^{n} \mu_Y = \mu_Y$$

$$var(\bar{Y}) = var(\frac{1}{n}\sum_{i=1}^{n} Y_i) = \frac{1}{n^2} var(\sum_{i=1}^{n} Y_i)$$

$$= \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n} cov(Y_i, Y_j) = \frac{1}{n^2}\sum_{i=1}^{n} var(Y_i)$$

$$= \frac{\sigma_Y^2}{n} \rightarrow 0 \text{ as } n \rightarrow \infty \text{ (i.i.d.)}$$

Review of Statistics

- The variance of $\bar{Y}$ is inversely proportional to n
- The sampling uncertainty associated with $\bar{Y}$ is proportional to $1/\sqrt{n}$ (square-root law)

Central Limit Theorem (CLT): If $Y_1, \ldots, Y_n$ are i.i.d. and $0 < \sigma_Y^2 < \infty$, then $\bar{Y}$ is an asymptotically normal estimator of μ_Y, that is, the distribution function of $\bar{Y} - E[\bar{Y}]/\sqrt{var(\bar{Y})}$ converges to a normal distribution function as n goes to infinity. It is written as

$$\frac{\bar{Y} - E[\bar{Y}]}{\sqrt{var(\bar{Y})}} = \frac{\sqrt{n}(\bar{Y} - \mu_Y)}{\sigma_Y} \xrightarrow{d} N(0,1)\,,$$

for which we read " $\sqrt{n}(\bar{Y} - \mu_Y)/\sigma_Y$ converges in distribution to $N(0,1)$".

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The exact (finite sample) distribution of $\bar{Y}$ has mean μ_Y ($\bar{Y}$ is unbiased) and variance σ_Y^2/n. Other than its mean and variance, the exact distribution of $\bar{Y}$ is complicated (depends on n and the population distribution), but if n is large, the sampling distribution is simple!

- As n increases, the distribution of $\bar{Y}$ becomes more tightly centered around μ_Y (LLN). Yes, $\bar{Y}$ is close to μ_Y.

$$\bar{Y} \xrightarrow{p} \mu_Y$$

- Moreover, the distribution of $\bar{Y} - \mu_Y$ becomes normal (CLT). $\bar{Y}$ approaches to μ_Y with $\bar{Y} - \mu_Y$ appearing bell shaped.

$$\frac{\bar{Y} - E[\bar{Y}]}{\sqrt{var(\bar{Y})}} \xrightarrow{d} N(0,1)$$

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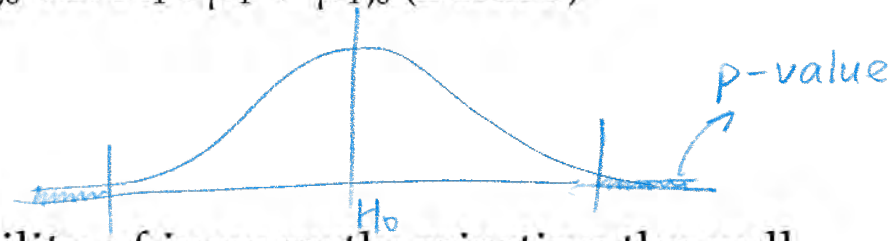
Hypothesis testing

- Make a decision whether a null hypothesis regarding some unknown parameter of the population distribution is true, or instead that some alternative hypothesis is true
  - For the papulation mean μ_Y, we can test

$$H_0 : \mu_Y = \mu_{Y,0} \text{ v.s. } H_1 : \mu_Y > \mu_{Y,0} \text{ (1-sided, } >)$$

$$H_0 : \mu_Y = \mu_{Y,0} \text{ v.s. } H_1 : \mu_Y < \mu_{Y,0} \text{ (1-sided, } <)$$

$$H_0 : \mu_Y = \mu_{Y,0} \text{ v.s. } H_1 : \mu_Y \neq \mu_{Y,0} \text{ (2-sided)}$$



Significance level

- A pre-specified probability of incorrectly rejecting the null, when the null is true. The probability of making type-I error
  - 1%, 5% and 10% are commonly used

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t test

- Consider the 2-sided hypothesis. It rejects the null if the t statistic $t = (\bar{Y} - \mu_{Y,0})/SE(\bar{Y})$ is too far away from zero, where $SE(\bar{Y})$ is the standard error of $\bar{Y}$
- Under the null, $t \xrightarrow{d} N(0,1)$. Thus, for a 5% test, the critical value is 1.96. That is, reject the null if $|t| > 1.96$

$SE(\bar{Y})$

- It is equal to $SE(\bar{Y}) = s_Y/\sqrt{n}$ where s_Y^2 is the sample variance

$$s_Y^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2$$

- By LLN and others, $s_Y^2 \xrightarrow{p} \sigma_Y^2$ and $\sqrt{n}SE(\bar{Y}) = s_Y \xrightarrow{p} \sigma_Y$

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p-value

- The probability of drawing a statistic (e.g. t statistic) at least as adverse to the null as the value actually computed with your data, assuming that the null hypothesis is true
  - Consider the 2-sided hypothesis. The p-value is

$$P(|t| > |t^{act}||H_0) = P(|N(0,1)| > |\bar{Y}^{act} - \mu_{Y,0}|/SE(\bar{Y})) ,$$

    where t^{act} and $\bar{Y}^{act}$ are the actual value of t and $\bar{Y}$

  - The t test is equivalent to rejecting the null if the p value is smaller than the significance level. The p-value is sometimes called the marginal significance level, and it contains more information than a "reject or not"

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Why is it called t test?

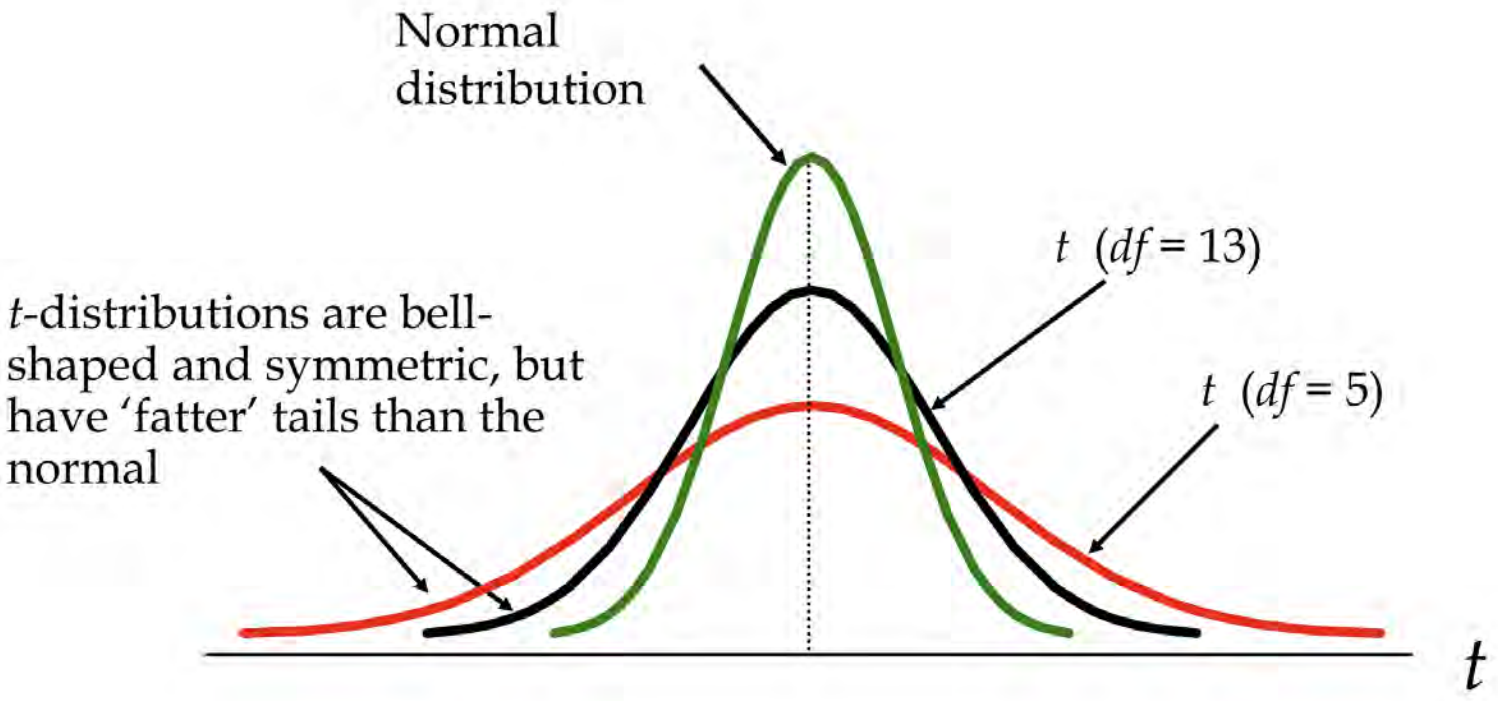
- So far the result is based on CLT when n is large
- If Y_i's are i.i.d. $N(\mu_Y, \sigma_Y^2)$, then the t statistic has the Student t distribution with degrees of freedom $n-1$, denoted as t_{n-1}. Then, critical values should be from t_{n-1}, rather than $N(0,1)$
- The theory of the t distribution was one of the early triumphs of mathematical statistics. If Y_i's are i.i.d. normal, then you can know the exact finite sample distribution of the t statistic. It is a conceptually beautiful result. But the normality assumption may be not true in economic data
  - Do you think earnings are normally distributed?

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The ***t*-distribution** is a continuous probability distribution with the following properties:

- It is bell-shaped and symmetric around the mean
- The area under the curve is equal to 1.0
- The shape of the curve depends on the **degrees of freedom** (*df*), $df = n - 1$
- The *t*-distribution is flatter and wider than the normal distribution
- The critical score for the *t*-distribution is greater than the critical *z*-score for the same confidence level

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Review of Statistics

The *t*-distribution is actually a family of distributions. As the number of degrees of freedom increases, the shape of the *t*-distribution becomes similar to the normal distribution

- With more than 100 degrees of freedom (a sample size of more than 100), the two distributions are practically identical

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t_{n-1} v.s. $N(0,1)$

- In fact, the difference between the t_{n-1} and N(0,1) critical values is negligible when n is large

Degrees of Freedom (n-1)	10	20	30	60	∞
5% t Distribution Critical Value	2.23	2.09	2.04	2.00	1.96

- Statistical software typically uses the t distribution to compute p values - but this is irrelevant when the sample size is moderate or large ($n \geq 30$)
- You can challenge the result when n is too small, e.g. $n = 10$. In this course, we will focus on the large–n approximation given by CLT

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Confidence interval

- Construct an interval using the data that contains the true value of some unknown parameter of the population distribution with a pre-specified confidence level. Lower and upper bounds are function of $Y_1, \ldots, Y_n$
  - Commonly used confidence levels are 99%, 95% and 90%
  - Also called interval estimation (v.s. point estimation)

95% confidence interval for μ_Y

- Constructed by inverting the 5% two-sided t test:

$$\bar{Y} \pm 1.96 \times SE(\bar{Y}) = [\bar{Y} - 1.96 \times SE(\bar{Y}), \bar{Y} + 1.96 \times SE(\bar{Y})]$$

- It contains μ_Y with 95% probability. The interval is random and may differ from one sample to the next, and μ_Y is fixed.
- The 5% two-sided t test for H_0 is equivalent to checking whether the interval contains $\mu_{Y,0}$